

Supplementary Materials: Dictionarium Sinicum

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1 Sample Dictionary Entries

We present three sample entries spanning common vocabulary, concrete nouns, and culturally significant loanword sources. Each entry shows definitions from community dictionaries and LLM-generated glosses across 18 languages.

1.1 你好 [nǐ hǎo] — Common Greeting

Lang	Source	Definition
en	cedict	hello; hi
de	handedict	Hallo! / Guten Tag!
de	minimax	Hallo!; Guten Tag!
fr	cfdict	bonjour / salut
fr	minimax	bonjour; salut
es	chedicc	hola; ¿qué tal?
id	cidict	halo; hai
ja	jmdict	ニーハオ
ja	minimax	こんにちは
ko	minimax	안녕; 안녕하세요
ru	minimax	привет
vi	minimax	xin chào; chào
tl	minimax	hello; kamusta
sv	minimax	hej; hallå
nl	minimax	hallo; dag
pt	minimax	olá; oi
it	minimax	ciao; buongiorno
ar	minimax	أهلاً / مرحباً
th	minimax	สวัสดี
hi	minimax	नमस्ते; हेलो
fa	minimax	سلام

Table 1: Dictionary entry for 你好 *nǐ hǎo* (hello). All 18 target languages: community sources and MiniMax glosses spanning Latin, CJK, Cyrillic, Arabic, Thai, and Devanagari scripts.

Headword	Meaning	Cantonese	Hokkien
你好	hello	<i>nei5 hou2</i>	<i>lí hó</i>
银行	bank	<i>ngan4 hong4</i>	<i>gîn-hâng</i>
豆腐	tofu	<i>dau6 fu6</i>	<i>tāu-hū</i>

Table 2: Dialect forms for the three sample entries. Cantonese in Jyutping romanization, Hokkien in POJ.

1.2 银行 [yín háng] — Bank

Lang	Source	Definition
en	cedict	bank
en	wiktextextract	bank (financial institution); jewellery dealer
de	handedict	Bank (S, Wirtsch)
fr	cfdict	banque
id	cidict	bank
ja	jmdict	ぎんこう
ko	minimax	은행; 금융 기관
es	minimax	banco
ru	minimax	банк
vi	minimax	ngân hàng
tl	minimax	bangko
th	minimax	ธนาคาร
ar	minimax	بنك / مصرف
hi	minimax	बैंक

Table 3: Dictionary entry for 银行 (bank). Community sources + selected MiniMax languages.

1.3 豆腐 [dòu fu] — Tofu (Loanword Source)

This entry illustrates both the multilingual definitions and the dialect forms that enable loanword tracing across Southeast Asia.

Lang	Source	Definition
en	cedict	tofu; bean curd
de	handedict	Tofu, Doufu, Bohnenkäse (S, Ess)
fr	cfdict	tofu
id	cidict	tahu; kembang tahu
ja	jmdict	とうふ
ko	minimax	두부
tl	minimax	tokwa
vi	minimax	đậu phụ
th	minimax	เต้าหู้

Table 4: Dictionary entry for 豆腐 (tofu).

The Hokkien form *tāu-hū* (Table 2) is the etymological source for Indonesian *tahu*, Tagalog *tokwa* (via Hokkien *tāu-koa*ⁿ “dried tofu”), and Thai เต้าหู้—all visible in the multilingual definitions above.

System	Sense coverage (%)				False sense rate (%)				Format compliance (%)			
	n	de	fr	id	n	de	fr	id	n	de	fr	id
Our pipeline	453	79.2	84.0	93.7	453	19.8	14.2	6.3	453	~100	~100	~100
MiniMax (generic)	177	58.3	67.8	83.6	177	45.4	29.1	17.6	177	97.9	86.4	89.5
Claude Sonnet 4	275	61.1	64.6	87.9	275	47.2	37.0	15.2	275	79.5	71.4	74.1
GPT-4o-mini	275	52.2	61.9	83.8	275	52.9	41.3	17.4	275	78.1	73.0	72.7
Google Translate	275	52.4	60.1	85.5	275	51.6	36.5	13.5	275	76.7	74.6	68.3
DeepL	275	49.3	52.4	83.9	275	54.4	47.2	15.7	275	75.3	68.3	68.3
Azure Translator	275	48.7	60.3	86.4	275	53.0	39.1	12.9	275	72.6	68.3	68.3

Table 5: Per-language MT baseline breakdown across three metrics. Our pipeline’s advantage is most pronounced on German, where baselines score 20–30 points lower on sense coverage. Format compliance measures adherence to CEDICT-style gloss lists.

Back-translation evaluation (%)				LLM-as-judge scores (1–5)				
Language	n	Sense	False	Language	n	Acc.	Nat.	Comp.
Spanish	126	79.0	20.2	Russian	6	4.75	4.50	3.75
Portuguese	123	75.2	22.3	Arabic	3	4.67	4.50	4.00
Italian	122	74.7	25.4	Spanish	8	4.56	4.44	4.00
Swedish	118	74.2	24.2	Dutch	12	4.54	4.58	3.58
Russian	123	71.3	29.4	Portuguese	10	4.50	4.50	3.35
Dutch	125	69.7	28.2	Vietnamese	10	4.50	4.30	3.60
Persian	128	67.6	31.6	Swedish	6	4.42	4.58	3.50
Thai	113	66.8	31.4	Persian	8	4.25	4.56	3.00
Vietnamese	107	65.4	28.2	Korean	6	4.25	4.33	2.92
Arabic	129	65.3	34.7	Italian	6	4.00	4.08	2.92
Hindi	128	63.4	32.3	Thai	7	4.07	4.36	3.14
Tagalog	110	61.7	34.1	Hindi	13	4.04	3.77	3.04
Korean	119	60.0	38.2	Tagalog	5	3.90	3.60	3.60
Overall	1,571	68.9	29.2	Overall	100	4.34	4.31	3.38

Table 6: Evaluation of 13 non-community languages. Left: back-translation sense coverage. Right: LLM-as-judge quality scores (accuracy, naturalness, completeness). Judges: Claude Sonnet 4, Kimi K2, GPT-OSS-120B, DeepSeek V3.1. Mean pairwise inter-judge disagreement: 0.64 points.

Script	Purpose
eval_community_comparison.py	Core evaluation against 4 community dictionaries (de, fr, id). Shared stratified sampling and metrics.
eval_mt_baseline.py	MT baselines: MiniMax generic prompt, Google Translate, DeepL, Azure, Claude Sonnet 4, GPT-4o-mini.
eval_multi_model.py	Free LLM comparison via Groq, Cerebras, and SambaNova inference providers.
eval_backtranslation.py	Back-translation round-trip + LLM-as-judge quality scoring for 13 non-community languages.

Table 7: Evaluation script manifest. All require Python 3.11+ and a populated dictmaster.db. Results in eval-results/ (JSON).

2 Evaluation Details

Tables 5–7 present the full evaluation results and script manifest. All raw results are provided as JSON in eval-results/.

Sampling. Headwords are stratified by frequency (high: ≥ 100 articles, mid: 10–99, low: 1–9, rare: 0), 50 per band, seed=42, yielding ~ 275 evaluation pairs across de/fr/id.

Reproducibility. MiniMax M2.5 (abab7-chat-preview) via api.minimax.chat; \$0.30/M in-

put, \$1.20/M output. Database: 428K headwords $\times$ 18 languages in SQLite. MT baselines: Google Translate v2, DeepL Free, Azure Translator v3, Claude Sonnet 4, GPT-4o-mini; free LLMs via Groq, Cerebras, SambaNova.